

$\exists \text{violinplayer}(x)$

$\exists x(\text{violinplayer}(x))$

LANGUAGES FOR AI, MODULE 2
FEBRUARY 14, 2024

Time: 2 hours.

Please write the answers to exercises 1-2, 3-4 and 5-6 on three different sheets of paper.

1. (5 points) Consider the language of propositional logic. Use natural deduction to prove that the following holds, or find a counter-example to show that it does not hold (remember that $\neg F$ is only a shorthand for $F \rightarrow \perp$).

- $\vdash ((A \wedge B) \rightarrow (A \wedge B \wedge C)) \rightarrow C$
- $\vdash ((A \rightarrow C) \wedge (A \rightarrow B) \wedge A) \rightarrow (B \wedge C)$

$\forall y(\text{nurse}(y) \wedge \exists x(\text{violinplayer}(x)) \rightarrow \dots$

2. (5 points) Transform the following propositional logic formula into an equivalent formula in Conjunctive Normal Form

$$(A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow (C \rightarrow B))$$

3. (5 points) Bianchi, Rossi, and Neri are suspected of a crime. They testify as follows:

- Bianchi: "Rossi is guilty and Neri is innocent".
- Rossi: "If Bianchi is guilty then so is Neri".
- Neri: "I'm innocent, but at least one of the others is guilty".

Express the testimony of each suspect as a formula in propositional logic. Assuming that all testimonies are true, who is innocent and who is guilty?

4. (5 points) Formalise the following sentence into FOL:

- There is a violin player who is liked by every nurse.
- Some swans are not white.
- If someone is noisy everybody is annoyed.

$\exists x(\text{violinplayer}(x) \rightarrow \dots$

5. (6 points) Using MiniZinc or CLP, create a program that colors the following map keeping in mind that neighboring regions should not have the same color. For simplicity, you can use numbers instead of colors, e.g., 1 = red, 2 = blue, etc. Use the minimum number of different colors.

A	1	2	D	1	F
2	3	B			
C		1	E		

T V T
T C V T C
F V F

6. (6 points) Create a Prolog predicate `multiply(L, N, Res)` that multiplies the all elements of the given list L, N times. For example:

?- multiply([a, b, 2], 3, Res).
Res = [a, a, a, b, b, b, b, b, 2, 2, 2, 2].

$T \rightarrow F$

Note that you can create and call another predicate that uses an accumulator variable to keep count of how many times you have repeated the current element of the list.

$\text{multiply}([], 0, \text{Res}).$
 $\text{multiply}([_], N, [_]) :- \text{multiply}([], N-1, [_]).$

ES. 1

1) $((A \wedge B) \rightarrow (A \wedge B \wedge C)) \rightarrow C \stackrel{?}{=} 1$ so it can be false when A, B, C are all false

$$C \equiv 1$$

$$((A \wedge B) \rightarrow (A \wedge \overset{1}{B \wedge 1})) \equiv 1 \text{ so } A \wedge B \equiv 1 \quad A \equiv 1, B \equiv 1$$

2) $((A \rightarrow C) \wedge (A \rightarrow B) \wedge A) \rightarrow (B \wedge C) \stackrel{?}{=} 1$ so it is always true

$$B \wedge C \equiv 1$$

$$1 \wedge 1 \equiv 1$$

$$(A \rightarrow C) \wedge (A \rightarrow B) \wedge A \equiv 1$$

$$A \equiv 1$$

$$1 \rightarrow C \equiv 1$$

$$C \equiv 1$$

$$1 \rightarrow B \equiv 1$$

$$B \equiv 1$$

ES. 2

$$(A \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow (C \rightarrow B)) \stackrel{CNF}{=} 1(\neg A \vee (\neg B \vee C)) \vee (\neg A \vee (\neg C \vee B)) \equiv \neg(\neg A \vee \neg B \vee C) \vee (\neg A \vee B \vee \neg C) \equiv$$

$$\equiv (A \wedge B \wedge \neg C) \vee \neg A \vee B \vee \neg C \equiv \neg A \vee B \vee \neg C$$

$$B \vee (B \wedge \neg C) \equiv B$$

ES. 3

$$1) R \vee \neg N$$

$$2) B \rightarrow N$$

$$3) \neg N \wedge (R \vee B)$$

Assuming ① $\wedge$ ② $\wedge$ ③ $\equiv 1$

$$R \vee \neg N \equiv 1$$

$$B \rightarrow N \equiv 1$$

$$\text{so } B \rightarrow 1 \equiv 1$$

$$\text{so } B \equiv 1$$

$$\neg N \wedge (R \vee B) \equiv 1$$

$$\text{so } \underline{N \equiv 1}$$

$$\text{so } R \vee 1 \equiv 1$$

$$\text{so } \underline{R \equiv 1}$$

Bianchi and Neri are innocent while Rossi is guilty

ES. 4

$$1) \exists x : [\text{violin player}(x) \wedge (\forall y : \text{nurse}(y) \wedge \text{like}(y, x))]$$

$$2) \exists x : \text{swan}(x) \wedge \neg \text{white}(x)$$

$$3) \forall x (\text{noisy}(x) \rightarrow \forall y (\text{annoyed}(y)))$$

ES. 5

color (VARIABLES) :-

length(VARIABLES, 6),

VARIABLES = [A, B, C, D, E, F],

VARIABLES ins 1..3,

A #\= D, A #\= B, A #\= C,

B #\= D, B #\= E, B #\= C,

C #\= E,

D #\= E, D #\= F

MaxColor = max(VARIABLES)

labeling([min(MaxColor)])

BASE CONSTRAINT

↳ if they touch, they must have
different colours

} better to try
MINIMIZE

```
int: max_colors = 6;

var 1..max_colors: A;
var 1..max_colors: B;
var 1..max_colors: C;
var 1..max_colors: D;
var 1..max_colors: E;
var 1..max_colors: F;

constraint A != D;
constraint A != B;
constraint A != C;

constraint B != D;
constraint B != E;
constraint B != C;

constraint C != E;

constraint D != E;
constraint D != F;

var int: num_colors == max([A,B,C,D,E,F]);

solve minimize num_colors;
```

ES. 6

$\text{mul_el}(E, \tau, [E]) :- !.$

$\text{mul_el}(E, N, [E|Res]) :-$

$N_1 \text{ is } N - \tau,$

$\text{mul_el}(E, N_1, Res).$

$\text{multiply}([], 0, []) :- !.$

$\text{multiply}([H|T], N, RF) :-$

$\text{mul_el}(H, N, RH),$

$\text{multiply}(T, N, Res),$

$\text{append}(RH, Res, RF).$